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Lecture Notes «Matemātiskā analīze I»

CALCULUS I

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LECTURE 6.

DERIVATIVE II

Differentials and approximations

Limit $\lim_{t \to 0} \frac{sint}{t}$

Derivatives of the composite function (Chain rule)

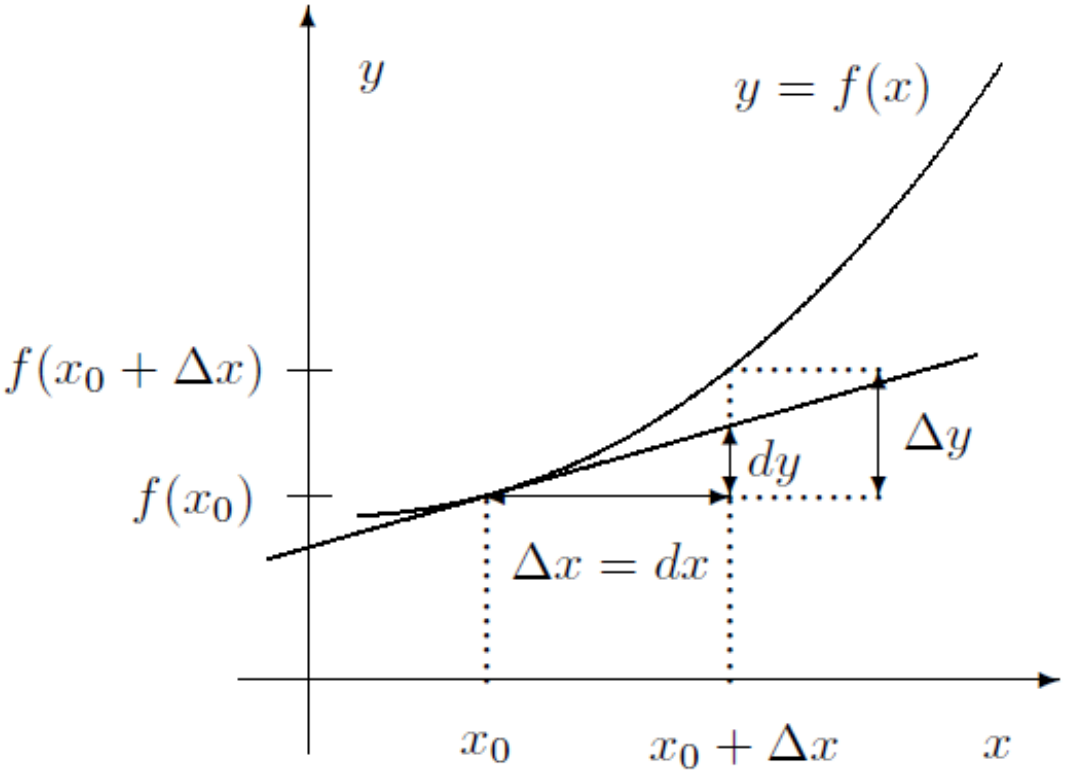
High-order derivatives

6. Homework

Example of the 1st tests

Differentials and approximations

Let $P(x_0, y_0)$ be a point on the graph of $y=f(x)$.
If a new coordinate system is created such that P is an origin, dx axis is parelel to the x axis and dy is paralel to the y axis. The equation of the tangent line at the point P is $dy = kdx = f'(x_0)dx$.



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Differentials and approximations

DEFINITION 6.1. Let assume that $\mathfrak{U}(x_0, y_0) \subseteq Dom(f)$. The function y=f(x) is differentiable at a point x_0, if at any neighbourhood $\mathfrak{U}(x_0, y_0)$ the increment Δy can be expressed as

$$\Delta y = a\Delta x + \varepsilon(\Delta x)\Delta x, where\ a \in \mathbb{R}\ and\ \varepsilon(0) = 0 = \lim_{\Delta x \to 0} \varepsilon(\Delta x).$$

The product $a\Delta x$ is called the differential of the function f(x) at the point x_0 and it is denotad by *dy*.

The increment Δx is denoted by *dx*, thus *dy = adx*.

Differentials and approximations

It can be proved that for one argument functions the existence of derivative and differentiability is equivalent statements. Therefore $a = f'(x)$. Later it will be shown that it is not true for functions with multiple amount of arguments.

Thus, $dy = f'(x_0)dx$ and therefore $f'(x_0) = \frac{dx}{dy}$. The derivate $f'(x_0) = \frac{dx}{dy}$can be interpreted as division of two differentials.

$\boldsymbol{dx}$ is called the differential of the independent variable x, it ies equal to $\Delta\boldsymbol{x}$, which is an arbitrary increment in the independant variable x.

$\Delta\boldsymbol{y}$ is the actual change in the variable y as x change from x to $x + \Delta x$, that is, $\Delta y = f(x + \Delta x) - f(x)$.
$\boldsymbol{dy} = f'(x)dx$ is called the differential of the dependent variable y or differential of the function $f(x)$.

Example

Please, find dy if $y = \sqrt{5x - 2}$.

Then $dy = d\left(\sqrt{5x - 2}\right) = \left(\sqrt{5x - 2}\right)' dx = \frac{5}{2\sqrt{5x-2}} dx$.

Differentials and approximations

Suppose that we have a function y = f(x). If argument x increases by Δx , then accordingly y increases by Δy that for a small changes is approximately equal to dy.
As $\Delta y = f(x + \Delta x) - f(x) \approx dy = f'(x_0)dx$, then

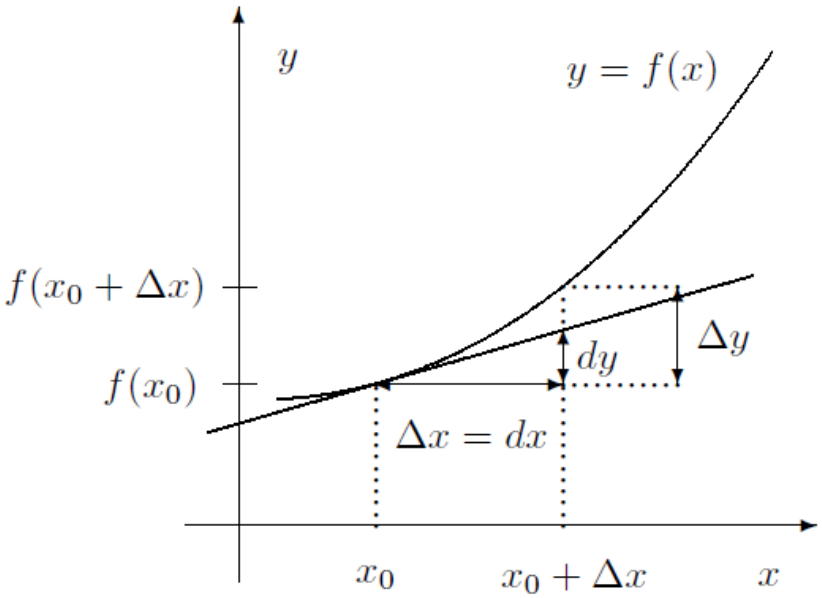
$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)dx.$$

Example

Please, find approximation of $\sqrt{16.8}$.

Here $x_0 = 16, \Delta x = 0.8, f(x) = \sqrt{x}\ and\ f'(x) = \frac{1}{2\sqrt{x}}$.

Thus, $\sqrt{16.8} \approx \sqrt{16} + \frac{1}{2\sqrt{16}} \cdot 0.8 = 4 + \frac{1}{8} \cdot 0.8 = 4 + 0.1 = 4.1$.



Limit $\lim_{t \to 0} \frac{sint}{t}$

One of the special trigonometric limits.
It is based on geometry and knowledge how to find the area of the triangles and sectors.

THEOREM 6.4.

$$\lim_{t \to 0} \frac{sint}{t} = 1$$

This limit is often used for removal of uncertainty $\frac{0}{0}$.

Limit $\lim\limits_{t\to 0}\frac{sint}{t}$

Examples

1. $\lim\limits_{x\to 0}\frac{\sin 9x}{x}=\lim\limits_{x\to 0}\frac{\sin 9x\cdot 9}{9x}=9.$

2. $\lim\limits_{t\to 0}\frac{1-\cos t}{t}=\lim\limits_{t\to 0}\frac{2\sin^2\frac{t}{2}}{t}=$

$$=\lim_{\substack{t\to 0\\(\frac{t}{2}\to 0)}}\frac{\sin\frac{t}{2}\sin\frac{t}{2}}{\frac{t}{2}}=\lim_{\substack{t\to 0\\(\frac{t}{2}\to 0)}}\frac{\sin\frac{t}{2}}{\frac{t}{2}}\lim_{t\to 0}\sin\frac{t}{2}=\sin 0=0.$$

3. $\lim\limits_{x\to 0}\frac{1-\cos^3 x}{x\sin 2x}=\lim\limits_{x\to 0}\frac{(1-\cos x)(1+\cos x+\cos^2 x)}{2x\sin x\cos x}=\lim\limits_{x\to 0}\frac{2\sin^2\frac{x}{2}\cdot 3}{2x\sin x\cdot 1}=$

$$=3\lim_{x\to 0}\frac{\sin^2\frac{x}{2}\cdot x}{4\cdot\left(\frac{x}{2}\right)^2\cdot\sin x}=\frac{3}{4}.\ \blacksquare$$

Exercises

Please, find the limits!

1) $\lim\limits_{x\to 0}\frac{\sin 4x}{\sin 16x}$

2) $\lim\limits_{x\to 0}\frac{\sqrt{3x+4}-2}{\sin 2x}$

Derivatives of the composite function (Chain Rule)

THEOREM 6.6.

Let y=f(u) and u=g(a). If g is differentiable at a and f is differentiable at g(a), then the composite function $f \circ g$, defined by $(f \circ g)(a) = f(g(a))$ is differentiable at *a* and

$$(f \circ g)'(a) = f'(b) \cdot g'(a),$$

where *b=g(a)*.

The derivative of a composite function is the derivative of the outer function evaluate at the inner function, times the derivative of the inner function.

Example

$y = (3x^2 - 8x + 1)^{100}$

$f(b) = b^{100}, b = g(x) = 3x^2 - 8x + 1$

$$((3x^2 - 8x + 1)^{100})' = 100(3x^2 - 8x + 1)^{99} \cdot (3x^2 - 8x + 1)' =$$

$$= 100(6x - 8)((3x^2 - 8x + 1)^{99}.$$

Derivatives of the composite function (Chain Rule)

Formulas for finding derivatives if argument is a function of variable *x* (composite function).

$$((u(x))^n)' = n(u(x))^{n-1} \cdot u'(x),\ n \in \mathbb{R},$$

$$(\sin u(x))' = \cos u(x) \cdot u'(x),$$

$$(\cos u(x))' = -\sin u(x) \cdot u'(x),$$

$$(\operatorname{tg} u(x))' = \frac{1}{\cos^2 u(x)} \cdot u'(x),$$

$$(\operatorname{ctg} u(x))' = -\frac{1}{\sin^2 u(x)} \cdot u'(x),$$

$$(a^{u(x)})' = a^{u(x)} \ln a \cdot u'(x),\ a > 0,$$

$$(e^{u(x)})' = e^{u(x)} \cdot u'(x),$$

$$(\log_a u(x))' = \frac{1}{u(x) \ln a} \cdot u'(x),\ a > 0,\ a \neq 1,$$

$$(\ln u(x))' = \frac{1}{u(x)} \cdot u'(x),$$

$$(\arcsin u(x))' = \frac{1}{\sqrt{1-u^2(x)}} \cdot u'(x),$$

$$(\arccos u(x))' = -\frac{1}{\sqrt{1-u^2(x)}} \cdot u'(x),$$

$$(\operatorname{arctg} u(x))' = \frac{1}{1+u^2(x)} \cdot u'(x),$$

$$(\operatorname{arcctg} u(x))' = -\frac{1}{1+u^2(x)} \cdot u'(x).$$

Derivatives of the composite function (Chain Rule)

Examples

1. $\left(\cos\frac{x^2+5}{x+1}\right)' = -\sin\frac{x^2+5}{x+1}\cdot\left(\frac{x^2+5}{x+1}\right)' = -\sin\frac{x^2+5}{x+1}\,\frac{2x(x+1)-(x^2+5)}{(x+1)^2} =$

$= -\frac{x^2+2x-5}{(x+1)^2}\sin\frac{x^2+5}{x+1}.$

2. If there are 3 or more functions – have to follow the same rule

$$\left(\ln\sin^3(4x-7)\right)' = \frac{1}{\sin^3(4x-7)}\left(\sin^3(4x-7)\right)' =$$

$$= \frac{1}{\sin^3(4x-7)}3\sin^2(4x-7)\left(\sin(4x-7)\right)' =$$

$$= \frac{3}{\sin(4x-7)}\cos(4x-7)\left(4x-7\right)' = \frac{12\cos(4x-7)}{\sin(4x-7)} = 12\mathrm{ctg}(4x-7). \blacksquare$$

Exercises

Please, find the derivatives!

1) $y = \sin\frac{x^2}{x^3+5}$ 2) $y = \mathrm{tg}^5(12x^2-5)$ 3) $y = \frac{\arcsin x}{\sqrt{1-x^2}} + \frac{1}{2}\ln\frac{1-x}{1+x}.$

High-order derivatives

The operation of ifferentiation takes a function *f* and produces a new fuction *f'*. If we now differentiate *f'*, we produce still another function, denoted by *f''* (read «f double prime») and call the second derivative of f. It in turn, may be differentiated, thereby producinf *f'''*, which is called the third derivative of f, and so on.
Thus, $f^{(n+1)} = (f^{(n)})', n = 1,2, \ldots$

The denotation for first derivative:

$$f'(x) = \frac{dy}{dx} = \frac{df(x)}{dx} = D_x y = D_x f(x)$$

Similarly:

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d^2f(x)}{dx^2} = D^2{}_x y = D^2{}_x f(x)$$

$$f'''(x) = \frac{d^3y}{dx^3} = \frac{d^3f(x)}{dx^3} = D^3{}_x y = D^3{}_x f(x)$$

$$\ldots$$

High-order derivatives

Example

If $f(x) = 4x^3 - 6x^2 - 9$, then
$f'(x) = 12x^2 - 12x$,
$f''(x) = 24x - 12$,
$f'''(x) = 24$,
$f^{IV}(x) = 0$.
As *0'=0*, then all other higher order derivatives are 0.

Exercises.

Please finf $f'''(x_0)$ if $f(x)=sin2x$, $x_0=0$.
Please finf $f^n(x)$ if $f(x) = \frac{1}{x}$.

HOMEWORK 6

11. Please, find limits by using rule $\lim_{t\to 0}\frac{sint}{t} = 1$.

a) $\lim_{x\to 0}\frac{sinx}{sin6x-sin7x}$

b) $\lim_{x\to 0}\frac{1-cosx}{sin^2 4x}$

Answers: a) -1 b) $\frac{1}{32}$

12. Please, find the derivatives and simplify.

a) $y = \ln(x^2 + \sqrt{x^4+1})$

b) $y = \frac{sin^2 x}{1+ctgx} + \frac{cos^2 x}{1+tgx}$

Answers: a) $\frac{2x}{\sqrt{x^4+1}}$ b) $-cos2x$ $(vai\ sin^2 x - cos^2 x)$

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p. 73 – 77, 118 – 130m 142-147.
Chapter 1. Section 1.4, Chapter 2. Sections 2.5.-2.6, 2.9.

EXAMPLE OF THE 1ST TEST

1. **Please, find limits!**

a) $\lim\limits_{x\to -2} \frac{x^2-2x-8}{2x^2+3x-2}$

b) $\lim\limits_{x\to 1} \frac{\sqrt{2}-\sqrt{x+1}}{1-\sqrt{3x-2}}$

c) $\lim\limits_{x\to 1^+} \frac{\lfloor x\rfloor+1}{\lfloor 5x\rfloor}$

d) $\lim\limits_{t\to 0} \frac{tg5t}{tg10t}$

2. **Please, find *a*** such that the function

$f(x) = \begin{cases} x^2+5, x<1 \\ 3x+a, x\geq 1 \end{cases}$ would be continuous $\forall x \in \mathbb{R}$. Please, justify the solution and sketch the graph with the found constant!

Please, sketch the graph if *a=0*! What kind of point of discontinuity exists in this case?

3. **Please, find the derivatives!**

a) $y = \left(4x^2 + \frac{1}{x^2}\right) lnx$

b) $y = \frac{5x^2-sinx}{cosx+2x^5}$

c) $y = (sin^3 2x + 4cos2x)^{10}$